

Validation Source Reliability Determination Module (VSRDM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VALIDOS® — Validation Governance Architecture
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· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
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Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation Source Reliability Determination

Canonical Definition

Validation Source Reliability Determination Module (VSRDM) defines the governance framework for integrating source identity, attribution, provenance, lineage, capability, competence, independence, conflicts, incentives, dependencies, transformations, limitations, uncertainty, corroboration, and source-role requirements into a formal determination of whether a Validation Source can be sufficiently relied upon for its assigned role within a governed validation process.

It establishes the final source-reliability governance gate between **Source Qualification** and subsequent **Validation Execution Governance** within VALIDOS®.

Operational Role

VSRDM serves as the fifth and final internal governance space of the Validation Source Reliability Standard.

The preceding modules establish:

VSIAM — Validation Source Identity & Attribution

VSPLM — Validation Source Provenance & Lineage VSCCM — Validation Source Capability & Competence

VSICM — Validation Source Independence & Conflict

VSRDM asks:

Considering the complete governed source relationship, can this source be sufficiently relied upon for the validation role assigned to it?

The governed progression is:

Identified Source → Traceable Source → Capable & Competent Source → Independence & Conflict Assessment → Integrated Reliability Assessment → Source Reliability Determination

VSRDM does not determine whether Validation Evidence is fit.

It does not determine whether the Validation Object is valid.

It determines whether the complete source relationship supporting Validation Evidence is sufficiently reliable for its assigned validation role.

Module Operational Space

VSRDM governs:

- integrated source reliability,
- source-reliability determination,
- source-reliability status,
- source-role reliability,
- reliability materiality,
- provenance reliability,
- capability contribution,
- competence contribution,
- independence contribution,
- conflict contribution,
- dependency contribution,
- transformation contribution,
- corroboration contribution,
- reliability limitations,
- reliability uncertainty,
- reliability conditions,
- reliability restrictions,
- source reliability readiness,
- source reliability expiration,
- source reliability reassessment,
- source reliability suspension,
- source reliability withdrawal,
- reliability continuity,
- and source-reliability traceability.

Module Function

VSRDM applies after the material source conditions have been assessed through the preceding VSRS modules.

Its function is to prevent:

- source reliability being reduced to reputation,
- perfect source identity being mistaken for reliability,
- strong competence overriding severe conflicts automatically,
- independence being treated as sufficient without capability,
- multiple source strengths being blindly averaged,
- unresolved provenance gaps disappearing during final assessment,
- conditional source relationships being represented as unconditional trust,
- corroboration being counted without accounting for shared lineage,
- source uncertainty being converted into confidence,
- and qualified sources being treated as globally reliable beyond the role actually assessed.

Source Reliability as an Integrated Condition

VALIDOS® treats Source Reliability as an integrated governance condition.

Reliability cannot be established from one characteristic alone.

VSRDM considers the complete source structure across:

Identity & Attribution

Provenance & Lineage

Capability & Competence

Independence & Conflict

together with material:

Dependencies

Transformations

Corroboration

Limitations

Uncertainty

and:

Assigned Source Role

A source may perform strongly across several dimensions while remaining unreliable for one critical evidentiary role because of a material weakness elsewhere.

Reliability Integration Principle

VSRDM establishes:

Source reliability should not be determined through blind averaging.

For example:

Strong competence cannot automatically neutralize an uncontrolled conflict.

Perfect independence cannot compensate for lack of capability.

Clear source identity cannot compensate for unknown provenance where provenance is critical.

Multiple favorable reports cannot compensate for one shared compromised upstream source.

Reliability determination must therefore remain materiality-sensitive.

Source Reliability Inputs

VSRDM may consume governed outputs including:

- Source Identity Records,
- Evidence Attribution Records,
- Provenance Records,
- Source Lineage Maps,
- Provenance Gap Records,
- Source Capability Assessments,
- Source Competence Assessments,
- Source Role Fitness Determinations,
- Source Independence Assessments,
- Conflict-of-Interest Records,
- Dependency Maps,
- Source Incentive Records,
- Corroboration Independence Assessments,
- transformation records,
- limitations,
- uncertainty,
- Validation Purpose,
- Validation Scope,
- Validation Depth,
- Validation Criteria,
- and assigned Source Role.

These inputs form the basis of the integrated Source

Source Role Reliability

Reliability is always assessed relative to a specific source role.

A source may therefore be:

Reliable as a measurement source

while:

Conditionally Reliable as an interpreter

and:

Unreliable as an independent assurance source

at the same time.

VALIDOS® establishes:

Source Reliability is role-specific, not universal.

Reliability Materiality

Not every source weakness has equal significance.

VSRDM may distinguish conditions that are:

- immaterial,
- minor,
- material,
- critical,
- or validation-blocking.

Materiality may depend upon:

- the source role,
- evidence criticality,
- Validation Criterion affected,
- Validation Depth,
- intended reliance,
- consequence of error,
- source substitutability,
- and availability of independent corroboration.

A critical source weakness must not disappear because the overall source profile is otherwise strong.

Critical Source Role

Certain source roles may be critical to validation.

Examples may include:

- sole measurement source,
- primary evidence origin,
- only available expert source,
- critical benchmark provider,
- sole operational record system,
- or independent assurance source.

Where a critical source is materially unreliable, downstream validation may require:

- replacement,
- additional corroboration,
- reduced scope,
- restricted reliance,
- or further governance action.

Provenance Reliability Contribution

Provenance and lineage affect reliability by revealing:

- origin clarity,
- upstream dependencies,
- transformation history,
- shared-source concentration,
- hidden intermediaries,
- or provenance gaps.

A source may be otherwise competent and independent while still carrying substantial reliability uncertainty because its evidentiary origin cannot be sufficiently reconstructed.

Capability & Competence Contribution

Capability and competence contribute to reliability only within the role and boundary in which they were established.

A source should not receive reliability credit beyond:

- its technical capability,
- operating range,
- professional competence,
- domain competence,
- informational access,

- or demonstrated task fitness.

Independence Contribution

Independence may strengthen source reliability where the source is expected to provide unbiased or independent evidence.

However:

Independent ≠ Automatically Reliable

Independence is one reliability dimension.

Its significance depends upon the source role.

Conflict Contribution

A conflict does not automatically produce Source Unreliable status.

VSRDM evaluates:

- conflict type,
- materiality,
- transparency,
- controls,
- residual conflict,
- available corroboration,
- and the role of the source.

A controlled minor conflict may coexist with strong reliability.

An uncontrolled material conflict in a sole critical source may materially constrain reliability.

Dependency Contribution

A source may depend upon:

- external data,
- common methods,
- shared infrastructure,
- upstream sources,
- analytical tools,
- or other systems.

VSRDM evaluates whether these dependencies materially weaken source reliability or reduce the independence of corroboration.

Transformation Contribution

Evidence transformations may create reliability conditions.

Where a source:

- aggregates,
- translates,
- normalizes,
- filters,
- summarizes,
- classifies,
- or analytically transforms evidence,

VSRDM considers whether the transformation pathway is sufficiently governed for the assigned role.

Corroboration Contribution

Corroboration may strengthen source reliability confidence where genuinely independent sources support compatible conclusions.

However:

Corroboration Strength = Agreement × Independence × Lineage Diversity

conceptually, not merely source count.

VSRDM therefore consumes the independence and lineage findings established earlier rather than counting sources mechanically.

Reliability Limitation

A source may be reliable while carrying limitations.

Examples include:

- reliable only within one measurement range,
- reliable only for one population,
- reliable only for raw data generation,
- reliable only under current calibration,
- reliable only within one system version,
- or reliable only with independent corroboration.

Limitations remain part of the Reliability Determination.

Reliability Condition

A reliability condition specifies what must remain true for the Source Reliability status to remain applicable.

Examples include:

- continued calibration,
- unchanged model version,
- continued independence,
- defined contractual controls,
- preserved information access,
- ongoing certification,
- or continued availability of an independent corroborating source.

Conditions must remain visible throughout subsequent validation governance.

Reliability Restriction

A source may be restricted to a limited validation role.

For example:

May support evidence generation but not independent assurance.

or:

May support contextual interpretation but not final technical determination.

Role restriction allows useful sources to remain within validation without assigning them authority or confidence beyond their legitimate function.

Reliability Uncertainty

Where material source conditions remain unresolved, VSRDM preserves uncertainty.

Sources may therefore receive:

Source Reliability Undetermined

rather than being forced into binary trusted/untrusted classifications.

Uncertainty may arise from:

- incomplete provenance,
 - uncertain source identity,
 - competence ambiguity,
 - hidden dependencies,
 - unresolved conflicts,
 - or insufficient independence evidence.
-

Source Reliability Status

VSRDM may establish the following canonical source-reliability states:

Source Reliable

The source is sufficiently identifiable, traceable, capable, competent, appropriately independent, conflict-governed, and reliable for the defined source role.

Source Conditionally Reliable

The source may perform the assigned validation role only under explicit conditions, limitations, restrictions, corroboration requirements, or other controls.

Source Reliability Undetermined

Available information is insufficient to establish a defensible Source Reliability Determination.

Source Reliability Insufficient

Material source-reliability requirements remain unresolved or inadequately supported.

Source Unreliable

The source cannot legitimately perform the proposed validation role.

These statuses apply to the source-role relationship.

They do not represent Evidence Fitness or the Validation State of the Validation Object.

Reliable Does Not Mean Correct

VALIDOS® establishes:

Source Reliable ≠ Every Source Output Correct

A reliable source may still make an error.

Reliability means that the source relationship is sufficiently trustworthy for the role assigned within validation.

The actual evidentiary content must still be governed through the wider validation process.

Unreliable Does Not Mean False

Likewise:

Source Unreliable ≠ Evidence Automatically False

Evidence from an unreliable source may occasionally be correct.

The reliability status concerns how much governed confidence the source relationship can support.

Independent evidence may later confirm the same underlying condition.

Source Reliability and Evidence Fitness

VSRDM preserves the separation:

Evidence Fitness → Is the evidence fit?

Source Reliability → Can the source behind it be relied upon?

A robust validation may require both.

Possible combinations include:

Evidence Fit + Source Reliable

Evidence Fit + Source Conditionally Reliable

Evidence Fit + Source Reliability Undetermined

Evidence Unfit + Source Reliable

and other combinations.

No single combination automatically produces the final Validation Determination.

Reliability-Qualified Evidence Basis

Once source reliability has been established, the evidence basis can become a **Reliability-Qualified Evidence Basis**.

This does not mean that evidence has become true or that the Validation Object has passed validation.

It means that the evidentiary basis now carries both:

Evidence Fitness information

and:

Source Reliability information

necessary for subsequent validation execution.

Source Reliability Readiness

A source relationship may be sufficiently ready for subsequent validation governance when:

- source identity is established,
 - attribution is sufficient,
 - relevant provenance is available,
 - critical lineage dependencies are understood,
 - capability and competence are adequate,
 - independence requirements are addressed,
 - conflicts are governed,
 - limitations are explicit,
 - and an appropriate Source Reliability status exists.
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Conditional Progression

A Conditionally Reliable source may progress only where the relevant conditions can remain visible and enforceable.

Examples include:

- independent corroboration required,
- restricted source role,
- specific version only,
- use only within defined operating conditions,
- disclosure required,
- or enhanced oversight required.

These conditions must not disappear during Validation Execution.

Source Reliability Expiration

A Reliability Determination may expire or require reassessment because of:

- elapsed time,
 - calibration expiration,
 - qualification expiration,
 - model update,
 - system change,
 - ownership change,
 - organizational restructuring,
 - new conflict,
 - new upstream dependency,
 - provenance correction,
 - competence change,
 - or changed Validation Scope.
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Expiration means the previous determination can no longer be assumed current.

Source Reliability Reassessment

Possible reassessment outcomes include:

Reliability Maintained

Reliability Modified

Conditional Reliability

Additional Source Qualification Required

Reliability Suspended

Reliability Withdrawn

This prevents historical source trust from persisting after material source conditions have changed.

Reliability Suspension

A source may require temporary suspension where material concerns arise but cannot yet be fully resolved.

Examples include:

- suspected compromise,
- unresolved calibration issue,
- disputed provenance,
- newly discovered conflict,
- or pending competence review.

Suspension preserves uncertainty without prematurely issuing a final Unreliable determination.

Reliability Withdrawal

Reliability may be withdrawn where the basis of the previous determination no longer exists.

Possible triggers include:

- source manipulation,
 - proven provenance fraud,
 - critical capability failure,
 - uncontrolled conflict,
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- compromised system,
- materially false attribution,
- or another critical reliability failure.

Withdrawal must remain traceable.

Reliability Continuity

Source Reliability information should remain attached to the evidence throughout:

Validation Execution → Validation Determination → Validation State → Reliance & Revalidation

This preserves the ability to determine which sources supported a later validation result and what reliability conditions applied at the time.

Minimum Implementation Framework

1. Consolidate Source Reliability Inputs

Collect governed outputs from:

- Source Identity & Attribution,
- Source Provenance & Lineage,
- Source Capability & Competence,
- and Source Independence & Conflict.

2. Identify Material Reliability Conditions

Identify:

- limitations,
- uncertainties,
- dependencies,
- transformations,
- conflicts,
- independence conditions,
- corroboration requirements,
- and role restrictions.

3. Assess Materiality

Determine which conditions are:

- immaterial,
 - material,
-

- critical,
- or validation-blocking.

4. Assess Role-Specific Reliability

Evaluate the source specifically against the evidentiary role it is expected to perform.

5. Establish Reliability Status

Classify the source as:

- Source Reliable,
- Source Conditionally Reliable,
- Source Reliability Undetermined,
- Source Reliability Insufficient,
- or Source Unreliable.

6. Establish Conditions, Limitations, and Restrictions

Define the boundaries within which the Source Reliability status may legitimately apply.

7. Establish Reassessment Triggers

Identify material events requiring future review.

8. Create Source Reliability Record

Preserve:

- Validation Source reference,
- source identity,
- source role,
- attribution status,
- provenance status,
- lineage structure,
- capability status,
- competence status,
- independence status,
- conflict status,
- dependencies,
- transformations,
- corroboration information,
- limitations,
- uncertainty,
- restrictions,
- reliability status,
- determination basis,
- effective period,
- reassessment triggers,
- version,
- changes,

- and sufficient lifecycle traceability.

Source Reliability Governance Flow

VSRDM completes the VSRS internal progression:

Source Identity & Attribution → Source Provenance & Lineage → Source Capability & Competence → Source Independence & Conflict → Source Reliability Determination

This transforms an evidentiary source from an assumed origin into a formally governed reliability relationship.

Governance Outputs

VSRDM may produce:

- Source Reliability Determination,
- Source Reliability Record,
- Source Reliability Status,
- Reliability-Qualified Evidence Basis,
- Source Role Reliability Profile,
- Source Reliability Condition Record,
- Source Reliability Limitation Record,
- Source Reliability Restriction Record,
- Source Reliability Uncertainty Record,
- Source Reliability Reassessment Trigger,
- Source Reliability Expiration Condition,
- Source Reliability Suspension Record,
- Source Reliability Withdrawal Record,
- Source Reliability Change Record,
- and Source Reliability Continuity Reference.

Use Case 1 — External AI Safety Assessment

Scenario

An external consultancy performs an AI safety assessment.

The consultancy is technically competent, but the work is funded by the AI vendor and uses several datasets supplied by that vendor.

Application

VSRDM integrates:

- established source identity,
- clear provenance,
- strong competence,

- partial informational dependence,
- financial conflict,
- and available independent corroboration.

Result

The consultancy may receive:

Source Conditionally Reliable

for defined assessment roles, with funding and information dependencies preserved as explicit conditions rather than being ignored or treated as automatic disqualification.

Use Case 2 — Industrial Measurement Source

Scenario A calibrated industrial sensor provides critical measurements for validation.

The sensor is well identified, directly attributable, and technically capable.

A recent maintenance event creates uncertainty about calibration continuity.

Application

VSRDM integrates the otherwise strong reliability profile with the unresolved calibration condition.

Result

The source may receive:

Source Reliability Undetermined

or:

Source Conditionally Reliable

until the calibration condition is resolved.

Use Case 3 — Internet Claim Verification

Scenario

Dozens of online sources repeat the same factual claim.

Application

VSRDM consumes lineage analysis showing that most sources derive from one original report.

Several additional sources rely upon one shared secondary dataset.

Result

The apparent number of sources does not produce artificial reliability confidence.

The actual source structure is assessed according to origin, capability, independence, shared dependencies, and corroboration quality.

Architectural Position

Validation Source Reliability Determination is the fifth and final internal governance space of the **Validation Source Reliability Standard (VSRS)**.

The complete VSRS progression is:

Validation Source Identity & Attribution → Validation Source Provenance & Lineage → Validation Source Capability & Competence → Validation Source Independence & Conflict → Validation Source Reliability Determination

Together, these five governed spaces transform source trust from an assumption into a structured source-reliability determination.

The VALIDOS® lifecycle can now progress from:

Qualified Validation Evidence

to: **Reliability-Qualified Validation Evidence**

and then to the next governance question:

Was the approved validation process actually executed as governed?

Foundational Principle

A source becomes reliable for validation only when the complete source relationship supports the role it is expected to perform.

Canonical Closing Statement

Validation Source Reliability Determination Module completes source-reliability governance within VALIDOS® by integrating identity, attribution, provenance, lineage, capability, competence, independence, conflicts, incentives, dependencies, transformations, corroboration, limitations, uncertainty, conditions, restrictions, change, and traceability into a formal role-specific Source

Reliability Determination. VSRDM ensures that validation confidence is never derived from source reputation, authority, popularity, repetition, or apparent independence alone, but from a reconstructable and governable source relationship capable of legitimately supporting the evidentiary role upon which validation depends.

Parent Resources

→ [VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

→ [VALIDOS® Architecture Map](#)

→ [VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

→ [Previous module: Validation Source Independence & Conflict Module \(VSICM\)](#)

Related Documents

→ [Validation Source Reliability Standard](#)

→ [About Validation Source Reliability Standard](#)

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Canonical Source

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